CS-4982 Progress Report-1

Darin Thomson; 3776723; UNB

8th February, 2026

1 Progress Summary:

The project has been on-schedule based on the the timeline/ milestone table adopted during drafting the proposal. This includes the brain-storming of the topic/ discussion with my co-author on topics related to this research and exchanging information and key resources related to and surrounding the topic.

The drafting of the proposal had taken a little more time than expected. This increased the duration of drafting the proposal from 2 days (Jan 23rd - Jan 25th) to 3 days (Jan 22nd - Jan 25th).

The research process had begun as soon as the draft of the proposal was done and submitted. This process took longer than the allotted time and had gone over until Feb 5th instead. This caused the drafting of progress report - 1 to be moved to Feb 6th-8th. This will cause a delay on the first draft of the essay and the drafting of progress report - 2 slightly further than schedule. The remaining of the tasks should follow the timeline with a possibility of more tweaks/ modifications.

2 Deliverables Needed to Accomplish:

2.1 Researching Process

2.2 Start drafting the first/ initial draft of the essay

2.3 Draft of progress report - 2

2.4 Video to record as part of the progress report - 2

2.5 Explore other possible resources related and surrounding the research. This would possibly involve a collaboration with my co-author. Note: This has not been included as part of my updated GANTT chart as it will be part of my initial draft process

2.6 Essay final draft which would involve strongly refining the initial draft - 1 of the essay

2.7 Lastly, the final commitment to accomplish would be the submission of the report including all the required deliverables and successfully presenting the research report to Dr. Baker and my peers

3 Solution Architect Diagram:

4 Old vs Updated GANTT Chart:

5 Updated Unforeseen Challenges/ Unknown Issues:

A key main unforeseen challenge currently being faced is that the operational model such as the code and software aspects of the project seem to be a lot more time and study than anticipated. However, this will neither cause changes to the timeline or the GANTT chart as it will be accommodated within the drafting of the initial report and the progress report - 2.

6 Updated Project Evaluation Tasks:

The tasks required to be accomplished would remain the same as mentioned in the initial proposal. The current evaluation metrics:

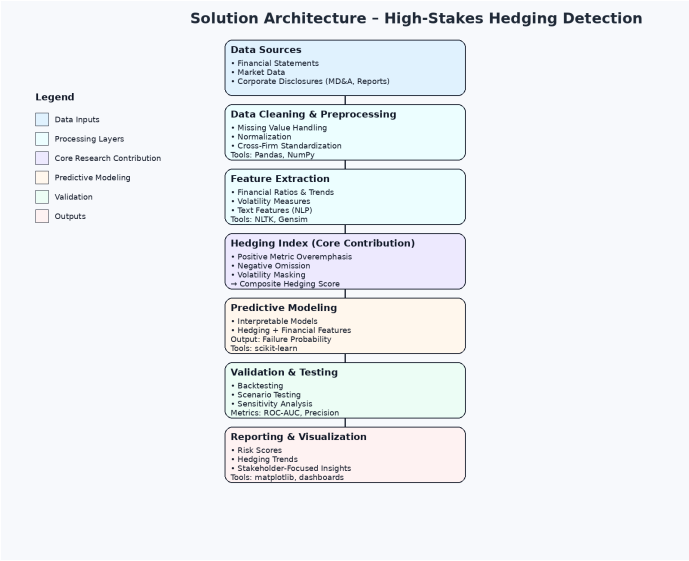

Figure 1: Solution Architect Diagram

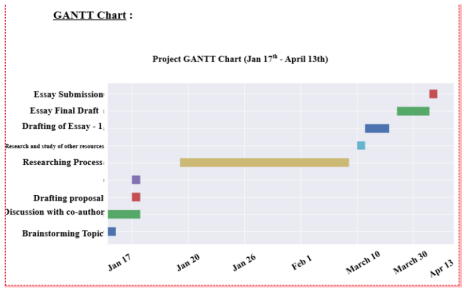

Figure 2: GANTT Chart

Figure 2: Old GANTT Chart

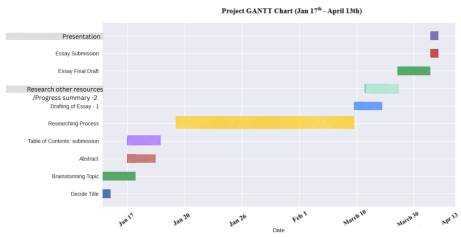

Figure 3: New GANTT Chart

6.0.1 How to test the prototype?

The prototype, when ready, could be tested using backtesting strategies, scenario-based testing and potentially using sensitivity analysis.

6.0.2 Success criteria of the Project

(a) If an accurate prediction of near-term company distress or failure is made using the research.

(b) Being able to identify statistically significant hedging behaviors of the companies considered.

(c) Robustness of our research/ approach across industries and time periods.

(d) Interpretable results that can potentially be used by our stakeholders as mentioned.

6.0.3 Metrics adopted

Hedging metrics (like overemphasis, volatility and hedging index), financial metrics (including PE ratios, debt-equity ratios, earnings ratios, etc.) and a few other validation metrics.

7 Updated References:

The references and key resources used during the process of this research study would also remain the same as far as progress report - 1. There may be po-

tential additions/ subtractions by the end of progress report - 2 which will be summarized later. Here are the current reference:

7.0.1 References that describe the problem statement

1. Loughran, T., and McDonald, B. (2024). Loughran-McDonald Master Dictionary with Sentiment Word Lists — Useful for explaining how corporate language can signal hedging or selective disclosure.

2. Loukas, L., Fergadiotis, M., Androutsopoulos, I., and Malakasiotis, P. (2021). EDGAR-Corpus: Billions of Tokens Make The World Go Round. ECONLP. — Provides context for large-scale analysis of corporate filings and textual disclosures.

3. Abney, S. (2007). Semi-Supervised Learning for Computational Linguistics. CRC Press. — Supports understanding the role of NLP in detecting hidden signals in text.

7.0.2 References that illustrate the Approach/ Methodology Adopted

4. Rehurek, R. (Gensim). Doc2Vec Tutorial — Shows how document embeddings can convert textual disclosures into numerical features for modeling.

5. Loukas et al., EDGAR-Corpus — Also supports methodology by providing structured large-scale data for model training.

6. Enhancing Financial Sentiment Analysis with BERT - Example of advanced NLP methods for predictive modeling of corporate disclosures.

7.0.3 References that report on resources I will adopt/ use

7. NetworkX Developers. NetworkX Documentation — For network-based feature analysis if needed.

8. Gensim / Doc2Vec Tutorial — For NLP and embeddings (also part of methodology, but primarily a resource/tool).

9. Loughran-McDonald Master Dictionary — Used as a lexical resource for sentiment/hedging detection.

10. EDGAR-Corpus dataset — Provides the textual data used as part of the process.

11. Python libraries like Pandas, NumPy, scikit-learn, matplotlib.

12. Abney, S. (2007) — Provides methodology guidance for NLP tools you will use.

8 Video (optional): Not Included (Will be done as part of progress report - 2)